A.2 Assignment 02 (2025)

A.2.1 Exercise 01

1. a. hello there:

h	e	l	l	o		t	h	e	r	e	NULL
68	65	6C	6C	6F	20	74	68	65	72	65	00

b. Cool

C	o	o	l	NULL
43	6F	6F	6C	00

c. boo!

b	o	o	!	NULL
62	6F	6F	21	00

d. bag o chips

b	a	g		o		c	h	i	p	s	NULL
62	61	67	20	6F	20	63	68	69	70	73	00

e. To the rescue!

T	o		t	h	e		r	e	s	c	u	e	!	NULL
54	6F	20	74	68	65	20	72	65	73	63	75	65	21	00

f. RISC-V

R	I	S	C	-	V	NULL
52	49	53	43	2D	56	00

Word Address	Data	Data	Data	Data	Data	Data
⋮	⋮	⋮	⋮	⋮	⋮	⋮
004F05CA					00	
004F05C9					21	
004F05C8					65	
004F05C7	0 0				75	
004F05C6	65				63	
004F05C5	72			00	73	
004F05C4	65			73	65	
004F05C3	68			70	72	
004F05C2	74			69	20	00
004F05C1	20			68	65	56
004F05C0	6F	00	00	63	68	2D
004F05BF	6C	6C	21	6F	74	43
004F05BE	6C	6F	6F	67	20	53
004F05BD	65	6F	6F	61	6F	49
004F05BC	68	43	62	62	54	52
	hello there	Cool	boo!	bag o chips	To the rescue!	RISC-V

2.

A.2.2 Exercise 02

1. nor

```
or s3, s4, s5
xori s3, s3, -1
```

2. nand

```
and s3, s4, s5
xori s3, s3, -1
```

A.2.3 Exercise 03

1. 0x50000000 overflow
2. 0xB0000000 no overflow
3. 0xD0000000 overflow

A.2.4 Exercise 04

1. There is an overflow if $128 + x6 > 2^{31} - 1$. In other words, if $x6 > 2^{31} - 129$. There is also an overflow if $128 + x6 < -2^{31}$. In other words, if $x6 < -2^{31} - 128$ (which is impossible given the range of x6).
2. There is an overflow if $128 - x6 > 2^{31} - 1$. In other words, if $x6 < 2^{31} + 129$. There is also an overflow if $128 - x6 < -2^{31}$. In other words, if $x6 > -2^{31} + 128$ x6 > 263 + 128 (which is impossible given the range of x6).
3. There is an overflow if $x6 - 128 > 2^{31} - 1$. In other words, if $x6 < 2^{31} + 127$ (which is impossible given the range of x6). There is also an overflow if $x6 - 128 < -2^{31}$. In other words, if $x6 < -2^{31} + 128$.

A.2.5 Exercise 05

1. 0xbabefef8
2. 0x23456780
3. 0x00000045

A.2.6 Exercise 06

1. 29

```
addi s7,zero, 29 # s7 = 29
```

2. -214

```
addi s7,zero, -214 # s7 = -214
```

3. -2999

```
lui s7, 0xFFFFF      # s7 = 0xFFFFF000
addi s7,s7, 0x449    # s7 = 0xFFFFF449 = -2999
```

4. 0xABCDE000

```
lui s7, 0xABCDE      # s7 = 0xABCDE000
```

5. 0xEDCBA123

```
lui s7, 0xEDCBA      # s7 = 0xEDCBA000
addi s7, s7, 0x123   # s7 = 0xEDCBA123
```

6. 0xEEEEEFAB

```
lui s7, 0xEEEEF      # s7 = 0xEEEEF
addi s7, s7, -85     # s7 = 0xEEEEEFAB
```

7. 47

```
addi s7,zero, 47 # s7 = 47
```

8. -349

```
addi s7,zero, -349 # s7 = -349
```

9. 5328

```
lui s7, 0x00001      # s7 = 0x00001000
addi s7,s7, 0x4D0    # s7 = 0x000014D0 = 5328
```

10. 0xBBCCD000

```
lui s7, 0xBBCCD      # s7 = 0xBBCCD000
```

11. 0xFEEBC789

```
lui s7, 0xFEEBC      # s7 = 0xFEEBC000
addi s7,s7, 0x789    # s7 = 0xFEEBC789
```

12. 0xCCAAB9AB

```
lui s7, 0xCCAAC      # s7 = 0xCCAAC000
addi s7,s7, -1621    # s7 = 0xCCAAB9AB = (-1621 = 0x9AB)
```

A.2.7 Exercise 07

```
xori x5, x6, -1
```